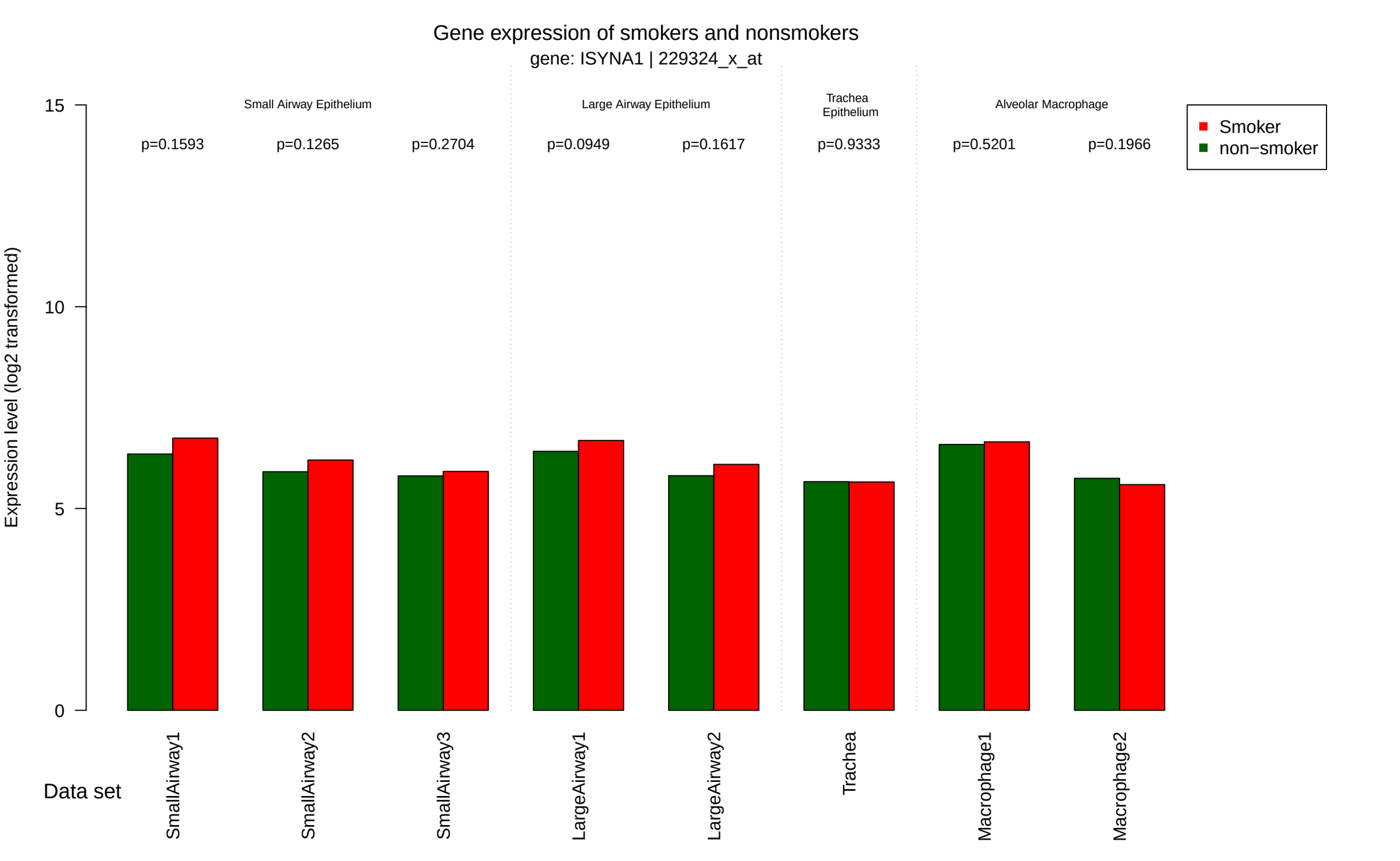
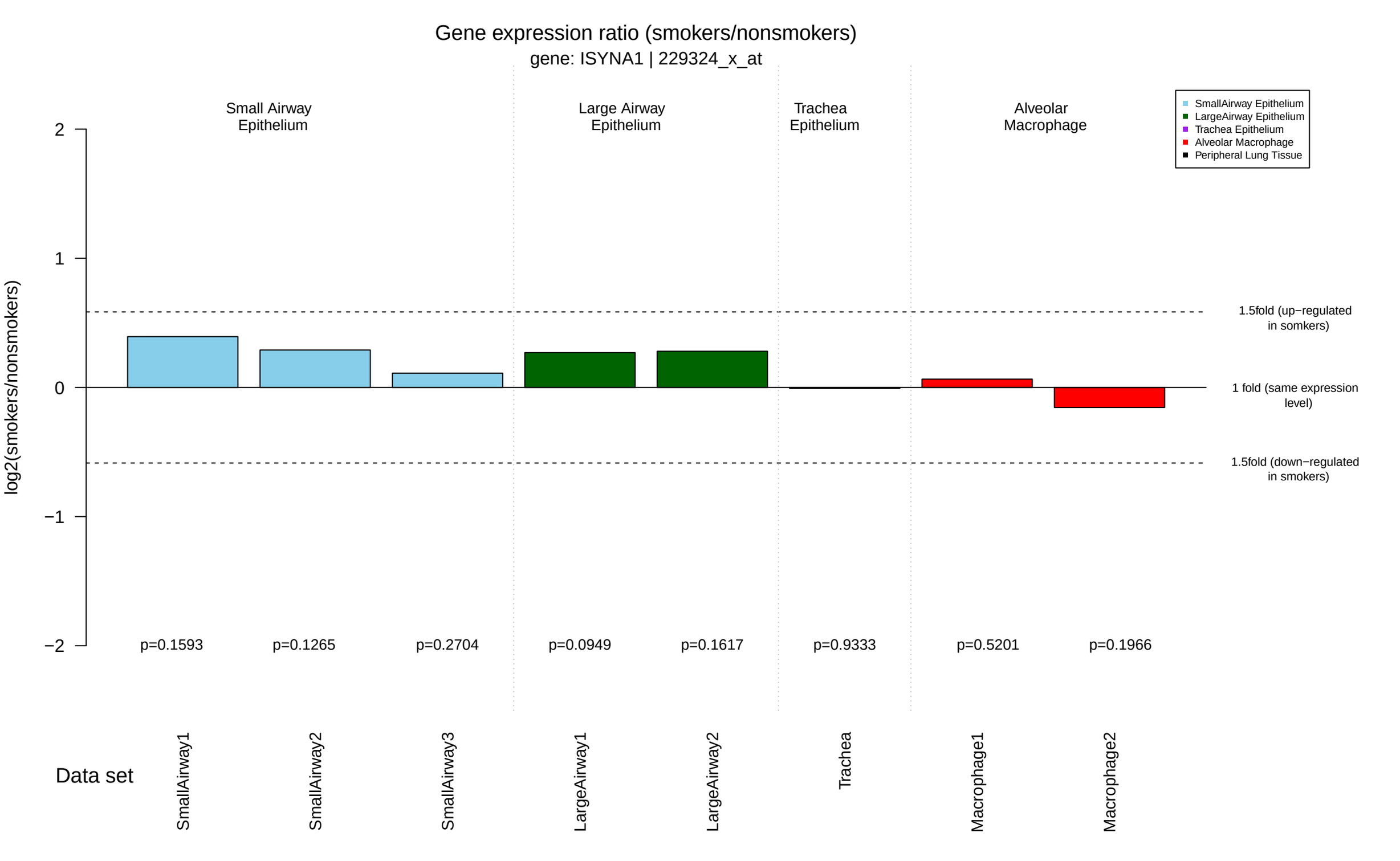
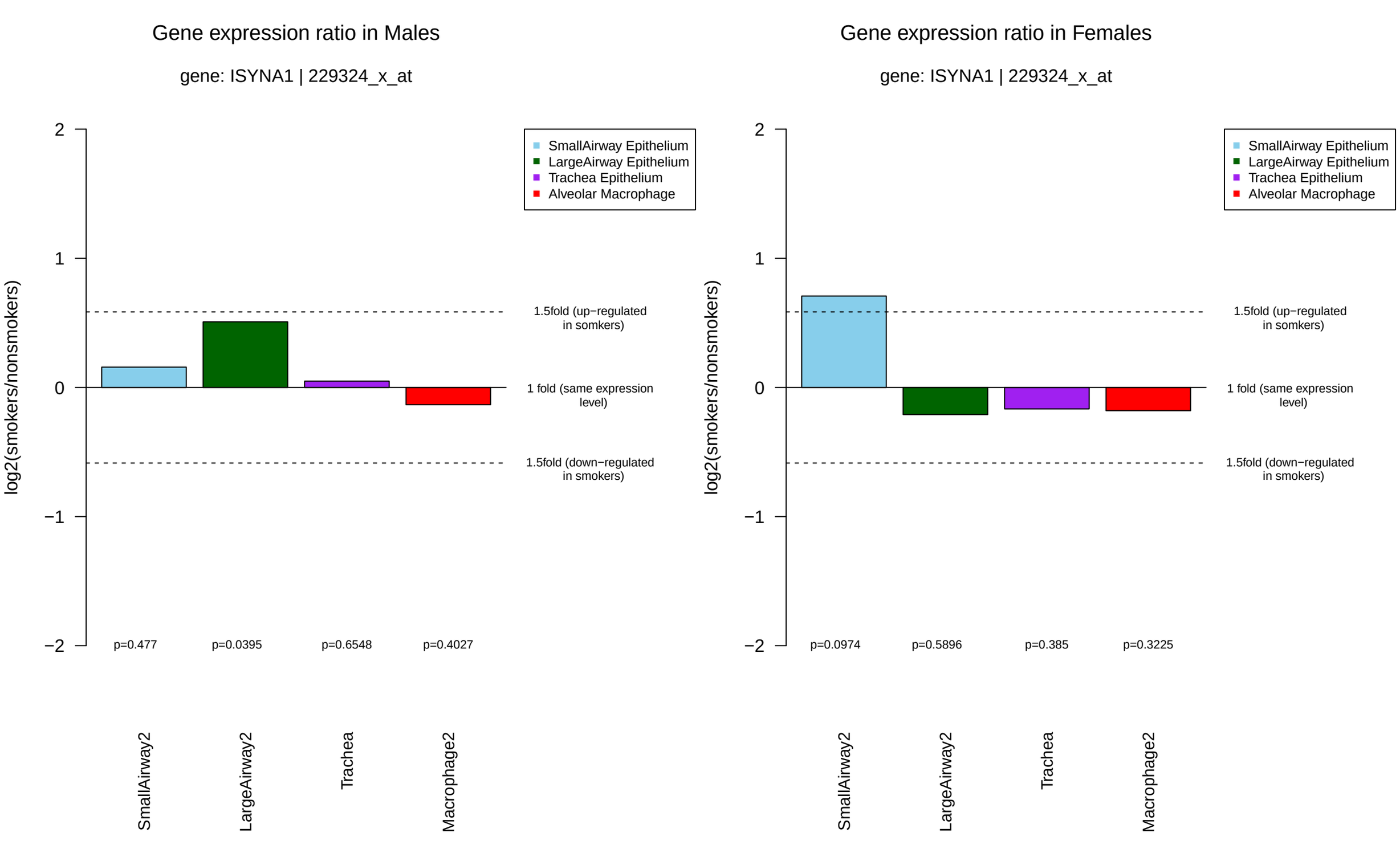
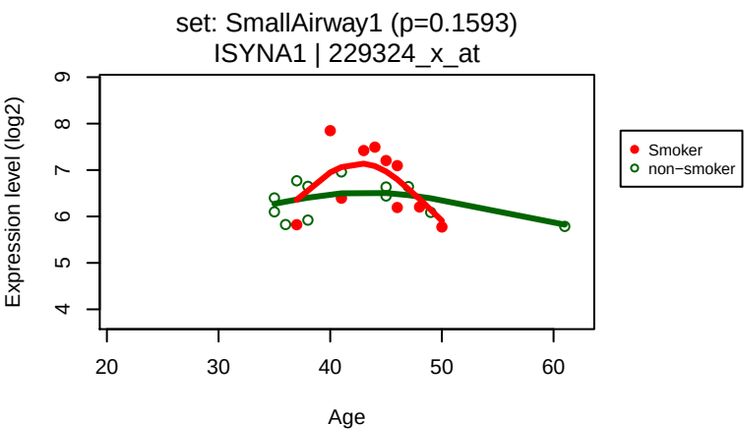
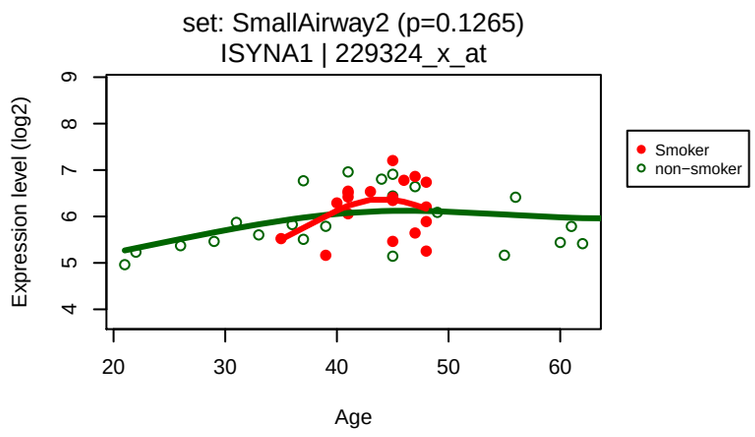
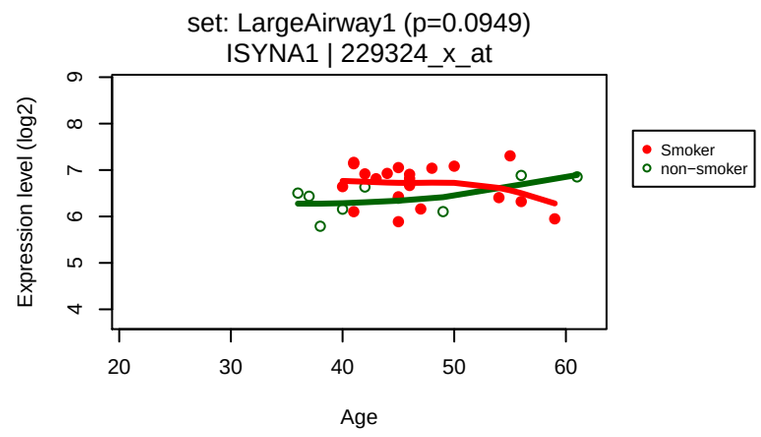

Gene expression of smokers and nonsmokers
gene: ISYNA1 | 229324_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1593
p=0.1265
p=0.2704
p=0.0949
p=0.1617
p=0.9333
p=0.5201
p=0.1966
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

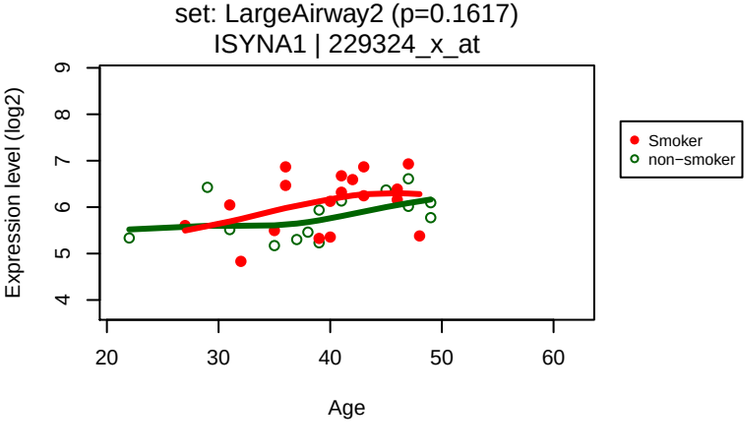

Gene expression ratio (smokers/nonsmokers)
gene: ISYNA1 | 229324_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1593
p=0.1265
p=0.2704
p=0.0949
p=0.1617
p=0.9333
p=0.5201
p=0.1966
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

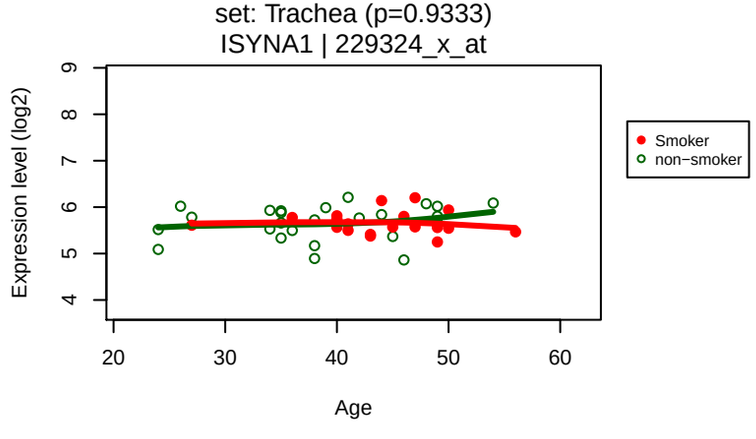

Gene expression ratio in Males
gene: ISYNA1 | 229324_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.477
p=0.0395
p=0.6548
p=0.4027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISYNA1 | 229324_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0974
p=0.5896
p=0.385
p=0.3225
SmallAirway2
LargeAirway2
Trachea
Macrophage2

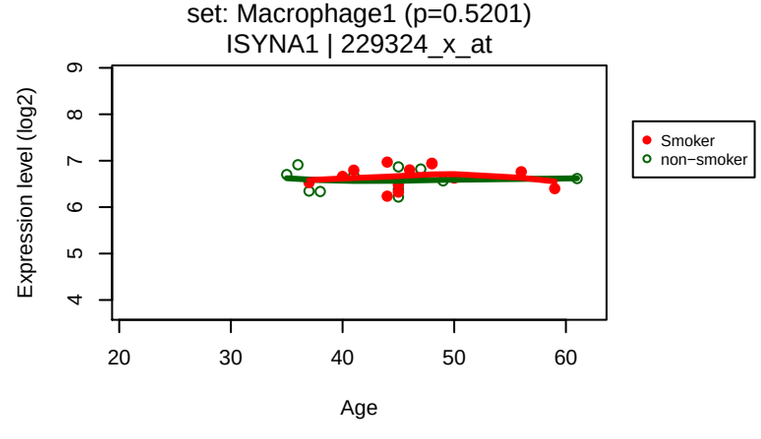

set: SmallAirway1 (p=0.1593)
ISYNA1 | 229324_x_at
Expression level (log2)
Age
Smoker
non−smoker

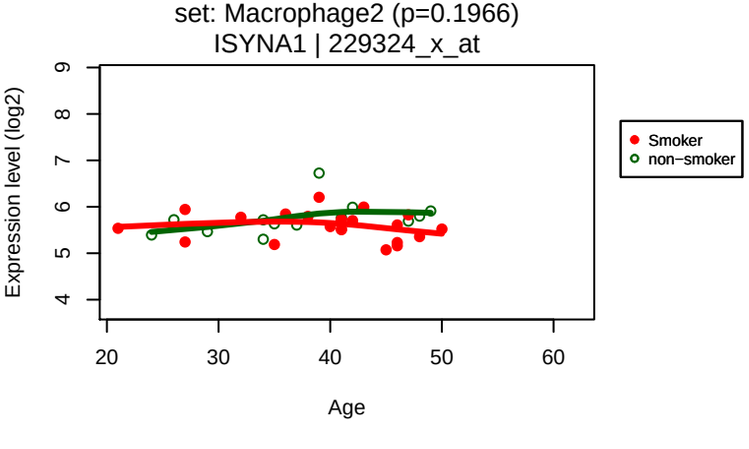

set: SmallAirway2 (p=0.1265)
ISYNA1 | 229324_x_at
Expression level (log2)
Age
Smoker
non−smoker

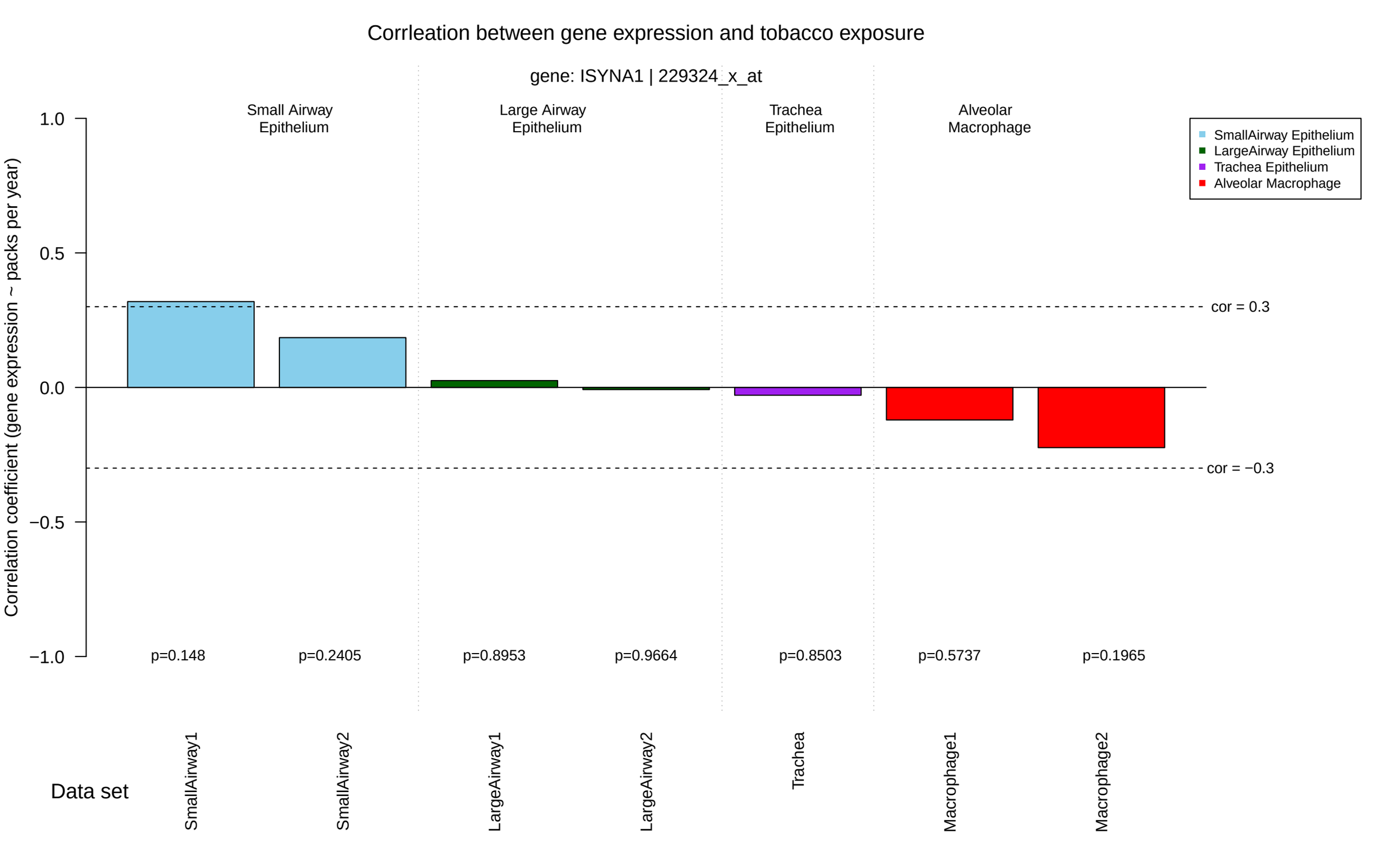

set: LargeAirway1 (p=0.0949)
ISYNA1 | 229324_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1617)
ISYNA1 | 229324_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9333)
ISYNA1 | 229324_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5201)
ISYNA1 | 229324_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1966)
ISYNA1 | 229324_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISYNA1 | 229324_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.148
p=0.2405
p=0.8953
p=0.9664
p=0.8503
p=0.5737
p=0.1965
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2